RENEWABLE ENERGY AND GREEN HYDROGEN

POWER METRIC FIREWALL → ENERGY CATEGORY MAP → GRID INTEGRATION RAIL → PROJECT STATUS LADDER → ELECTROLYSIS PATHWAY → HYDROGEN CLAIM STACK

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g2 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 POWER METRIC FIREWALL

POWER METRIC FIREWALL | INSTALLED CAPACITY | NAMEPLATE POWER CAPABILITY | ELECTRICAL ENERGY PRODUCED OVER TIME | CAPACITY SHARE | PORTFOLIO CAPABILITY, NOT OUTPUT SHARE | RELIABLE SUPPLY | TIMING, FLEXIBILITY AND NETWORK MATTER

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
INSTALLED CAPACITY → nameplate power capability	CAPACITY SHARE → portfolio capability, not output share	RULE → never convert megawatts into generation without evidence
GENERATION → electrical energy produced over time	RELIABLE SUPPLY → timing, flexibility and network matter	MUST REMEMBER: Renewable-energy transition must connect resource, installed capacity,...

CORE CLAIM
- INSTALLED CAPACITY → nameplate power capability
- GENERATION → electrical energy produced over time

MECHANISM
- CAPACITY SHARE → portfolio capability, not output share
- RELIABLE SUPPLY → timing, flexibility and network matter

CONSEQUENCE / LIMIT
- RULE → never convert megawatts into generation without evidence
- MUST REMEMBER: Renewable-energy transition must connect resource, installed capacity,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → never convert megawatts into generation without evidence.

01

STAGE 01 ENERGY CATEGORY MAP

ENERGY CATEGORY MAP | SOLAR WIND HYDRO BIOMASS GEOTHERMAL | RENEWABLE SOURCES PLUS OTHER NON-FOSSIL SOURCES | GENERATED ENERGY OUTPUT | MANUFACTURED ENERGY CARRIER | SOURCE, OUTPUT AND CARRIER ARE DIFFERENT CATEGORIES

RENEWABLE → solar wind hydro biomass geothermal

NON-FOSSIL → renewable sources plus other non-fossil sources → ELECTRICITY → generated energy output → HYDROGEN → manufactured energy carrier

CORE CLAIM
- RENEWABLE → solar wind hydro biomass geothermal
- NON-FOSSIL → renewable sources plus other non-fossil sources

MECHANISM
- ELECTRICITY → generated energy output
- HYDROGEN → manufactured energy carrier

CONSEQUENCE / LIMIT
- RULE → source, output and carrier are different categories
- RENEWABLE → solar wind hydro biomass geothermal

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → source, output and carrier are different categories.

02

STAGE 02 GRID INTEGRATION RAIL

GRID INTEGRATION RAIL | VARIABLE GENERATION | FORECAST RESOURCE-DEPENDENT OUTPUT | MOVE POWER FROM RESOURCE TO DEMAND | FLEXIBLE DEMAND AND BALANCING | MATCH SYSTEM CONDITIONS | SHIFT ENERGY AND PROVIDE SELECTED SERVICES | RELIABILITY IS A SYSTEM OUTCOME

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
VARIABLE GENERATION → forecast resource-dependent output	FLEXIBLE DEMAND AND BALANCING → match system conditions	DELIVERY → reliability is a system outcome
TRANSMISSION → move power from resource to demand	STORAGE → shift energy and provide selected services	VARIABLE GENERATION → forecast resource-dependent output

CORE CLAIM
- VARIABLE GENERATION → forecast resource-dependent output
- TRANSMISSION → move power from resource to demand

MECHANISM
- FLEXIBLE DEMAND AND BALANCING → match system conditions
- STORAGE → shift energy and provide selected services

CONSEQUENCE / LIMIT
- DELIVERY → reliability is a system outcome
- VARIABLE GENERATION → forecast resource-dependent output

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORAGE → shift energy and provide selected services.

03

STAGE 03 PROJECT STATUS LADDER

PROJECT STATUS LADDER | TARGET OR ANNOUNCEMENT | INTENDED FUTURE SCALE | ALLOCATION OR AWARD | CONTRACTUAL OR POLICY COMMITMENT | PHYSICAL IMPLEMENTATION STAGE | ASSET ENTERS SERVICE | GENERATION OR PRODUCTION

TARGET OR ANNOUNCEMENT → intended future scale

ALLOCATION OR AWARD → contractual or policy commitment → CONSTRUCTION → physical implementation stage → COMMISSIONING → asset enters service

CORE CLAIM
- VARIABLE GENERATION → forecast resource-dependent output
- TRANSMISSION → move power from resource to demand

MECHANISM
- FLEXIBLE DEMAND AND BALANCING → match system conditions
- STORAGE → shift energy and provide selected services

CONSEQUENCE / LIMIT
- DELIVERY → reliability is a system outcome
- VARIABLE GENERATION → forecast resource-dependent output

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: STORAGE → shift energy and provide selected services.

03

STAGE 03
PROJECT STATUS LADDER

PROJECT STATUS LADDER | TARGET OR ANNOUNCEMENT | INTENDED FUTURE SCALE | ALLOCATION OR AWARD | CONTRACTUAL OR POLICY COMMITMENT | PHYSICAL IMPLEMENTATION STAGE | ASSET ENTERS SERVICE | GENERATION OR PRODUCTION

TARGET OR ANNOUNCEMENT → intended future scale

ALLOCATION OR AWARD → contractual or policy commitment → CONSTRUCTION → physical implementation stage → COMMISSIONING → asset enters service

CORE CLAIM
- TARGET OR ANNOUNCEMENT → intended future scale
- ALLOCATION OR AWARD → contractual or policy commitment

MECHANISM
- CONSTRUCTION → physical implementation stage
- COMMISSIONING → asset enters service

CONSEQUENCE / LIMIT
- GENERATION OR PRODUCTION → measured operating output
- TARGET OR ANNOUNCEMENT → intended future scale

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ALLOCATION OR AWARD → contractual or policy commitment.

04

STAGE 04
ELECTROLYSIS PATHWAY

ELECTROLYSIS PATHWAY | ELECTRICITY SOURCE | ATTRIBUTE MUST BE EVIDENCED | WATER AND TREATMENT | PROJECT-SPECIFIC INPUT CONTEXT | WATER SPLIT INTO HYDROGEN AND OXYGEN | HYDROGEN OUTPUT | CHEMICAL PRODUCT

ELECTRICITY SOURCE → attribute must be evidenced → WATER AND TREATMENT → project-specific input context → ELECTROLYSER → water split into hydrogen and oxygen → HYDROGEN OUTPUT → chemical product → GREEN CLAIM → standard and certification layer still required

CORE CLAIM
- ELECTRICITY SOURCE → attribute must be evidenced
- WATER AND TREATMENT → project-specific input context

MECHANISM
- ELECTROLYSER → water split into hydrogen and oxygen
- HYDROGEN OUTPUT → chemical product

CONSEQUENCE / LIMIT
- GREEN CLAIM → standard and certification layer still required
- ELECTRICITY SOURCE → attribute must be evidenced

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GREEN CLAIM → standard and certification layer still required.

05

STAGE 05
HYDROGEN CLAIM STACK

HYDROGEN CLAIM STACK | PRODUCTION PATHWAY | HOW HYDROGEN IS MADE | LIFECYCLE BOUNDARY | INCLUDED EMISSIONS STAGES | OFFICIAL STANDARD | QUALIFYING CONDITIONS | DOCUMENTED CONFORMITY

PRODUCTION PATHWAY → how hydrogen is made → LIFECYCLE BOUNDARY → included emissions stages → OFFICIAL STANDARD → qualifying conditions → CERTIFICATION → documented conformity → MARKET LABEL → cannot replace the layers below

CORE CLAIM
- PRODUCTION PATHWAY → how hydrogen is made
- LIFECYCLE BOUNDARY → included emissions stages

MECHANISM
- OFFICIAL STANDARD → qualifying conditions
- CERTIFICATION → documented conformity

CONSEQUENCE / LIMIT
- MARKET LABEL → cannot replace the layers below
- PRODUCTION PATHWAY → how hydrogen is made

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MARKET LABEL → cannot replace the layers below.

06

STAGE 06
USE-CASE DECISION TREE

USE-CASE DECISION TREE | CAN CLEAN ELECTRICITY BE USED DIRECTLY | PREFER DIRECT ROUTE | MOLECULE OR HIGH HEAT NEEDED | TEST HYDROGEN FIT | SECTOR AND DUTY CYCLE | IDENTIFY TECHNICAL CONSTRAINT | CONVERSION LOSS AND INFRASTRUCTURE

CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route

MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit → SECTOR AND DUTY CYCLE → identify technical constraint → CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

CORE CLAIM
- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route
- MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit

MECHANISM
- SECTOR AND DUTY CYCLE → identify technical constraint
- CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

CONSEQUENCE / LIMIT
- VERDICT → hydrogen is selective, not universal
- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route

STAGE 06 USE-CASE DECISION TREE

USE-CASE DECISION TREE · CAN CLEAN ELECTRICITY BE USED DIRECTLY · PREFER DIRECT ROUTE · MOLECULE OR HIGH HEAT NEEDED · TEST HYDROGEN FIT · SECTOR AND DUTY CYCLE · IDENTIFY TECHNICAL CONSTRAINT · CONVERSION LOSS AND INFRASTRUCTURE

CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route

MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit → SECTOR AND DUTY CYCLE → identify technical constraint → CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

CORE CLAIM
- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route
- MOLECULE OR HIGH HEAT NEEDED → test hydrogen fit

MECHANISM
- SECTOR AND DUTY CYCLE → identify technical constraint
- CONVERSION LOSS AND INFRASTRUCTURE → disclose trade-off

CONSEQUENCE / LIMIT
- VERDICT → hydrogen is selective, not universal
- CAN CLEAN ELECTRICITY BE USED DIRECTLY → prefer direct route

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SECTOR AND DUTY CYCLE → identify technical constraint.

STAGE 07 DERIVATIVE CONVERSION MAP

DERIVATIVE CONVERSION MAP · BASE ENERGY CARRIER AND FEEDSTOCK · NITROGEN-BASED DERIVATIVE · METHANOL OR SYNTHETIC FUEL · CARBON-CONTAINING DERIVATIVE · CONVERSION AND DELIVERY · EXTRA ENERGY AND INFRASTRUCTURE · PRODUCT TARGETS AND OUTPUTS REMAIN SEPARATE

HYDROGEN → base energy carrier and feedstock

AMMONIA → nitrogen-based derivative → METHANOL OR SYNTHETIC FUEL → carbon-containing derivative → CONVERSION AND DELIVERY → extra energy and infrastructure

CORE CLAIM
- HYDROGEN → base energy carrier and feedstock
- AMMONIA → nitrogen-based derivative

MECHANISM
- METHANOL OR SYNTHETIC FUEL → carbon-containing derivative
- CONVERSION AND DELIVERY → extra energy and infrastructure

CONSEQUENCE / LIMIT
- RULE → product targets and outputs remain separate
- HYDROGEN → base energy carrier and feedstock

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → product targets and outputs remain separate.

STAGE 08 WATER AND SITE SCREEN

WATER AND SITE SCREEN · FRESHWATER TREATED WATER OR OTHER VERIFIED SUPPLY · PRETREATMENT REQUIREMENT · BASIN AND LOCAL SCARCITY CONTEXT · COMPETING DEMAND · ECOLOGICAL AND SOCIAL CLAIMS · PROJECT EVIDENCE, NOT A UNIVERSAL WATER NUMBER

SOURCE → freshwater treated water or other verified supply

QUALITY → pretreatment requirement → LOCATION → basin and local scarcity context → COMPETING DEMAND → ecological and social claims

CORE CLAIM
- SOURCE → freshwater treated water or other verified supply
- QUALITY → pretreatment requirement

MECHANISM
- LOCATION → basin and local scarcity context
- COMPETING DEMAND → ecological and social claims

CONSEQUENCE / LIMIT
- VERDICT → project evidence, not a universal water number
- SOURCE → freshwater treated water or other verified supply

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → project evidence, not a universal water number.

STAGE 09 HYDROGEN LOGISTICS CHAIN

HYDROGEN LOGISTICS CHAIN · HYDROGEN LEAVES ELECTROLYSER · COMPRESSION LIQUEFACTION OR CONVERSION · STORAGE AND TRANSPORT · VESSEL PIPELINE TANKER OR PORT · END USE · EQUIPMENT AND SAFETY COMPATIBILITY · DELIVERED SERVICE DIFFERS FROM PRODUCTION CAPACITY

PRODUCTION → hydrogen leaves electrolyser → CONDITIONING → compression liquefaction or conversion → STORAGE AND TRANSPORT → vessel pipeline tanker or port → END USE → equipment and safety compatibility → OUTCOME → delivered service differs from production capacity

CORE CLAIM
- PRODUCTION → hydrogen leaves electrolyser
- CONDITIONING → compression liquefaction or conversion

MECHANISM
- STORAGE AND TRANSPORT → vessel pipeline tanker or port
- END USE → equipment and safety compatibility

CONSEQUENCE / LIMIT
- OUTCOME → delivered service differs from production capacity
- CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free,.

STAGE 09 HYDROGEN LOGISTICS CHAIN

HYDROGEN LOGISTICS CHAIN · HYDROGEN LEAVES ELECTROLYSER · COMPRESSION LIQUEFACTION OR CONVERSION · STORAGE AND TRANSPORT · VESSEL PIPELINE TANKER OR PORT · END USE · EQUIPMENT AND SAFETY COMPATIBILITY · DELIVERED SERVICE DIFFERS FROM PRODUCTION CAPACITY

PRODUCTION → hydrogen leaves electrolyser → CONDITIONING → compression liquefaction or conversion → STORAGE AND TRANSPORT → vessel pipeline tanker or port → END USE → equipment and safety compatibility → OUTCOME → delivered service differs from production capacity

CORE CLAIM

- PRODUCTION → hydrogen leaves electrolyser
- CONDITIONING → compression liquefaction or conversion

MECHANISM

- STORAGE AND TRANSPORT → vessel pipeline tanker or port
- END USE → equipment and safety compatibility

CONSEQUENCE / LIMIT

- OUTCOME → delivered service differs from production capacity
- CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Capacity in MW is not generation in MWh, renewable is not impact-free,.

STAGE 10 MISSION AND SIGHT MAP

MISSION AND SIGHT MAP · MISSION TARGET · INTENDED FUTURE PRODUCTION SCALE · SIGHT MANUFACTURING · ELECTROLYSER CAPABILITY SUPPORT · SIGHT PRODUCTION · HYDROGEN OUTPUT SUPPORT · COMMISSIONED OUTPUT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
MISSION TARGET → intended future production scale	SIGHT PRODUCTION → hydrogen output support	CLIMATE OUTCOME → separate lifecycle result
SIGHT MANUFACTURING → electrolyser capability support	COMMISSIONED OUTPUT → later measured evidence	EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,...

CORE CLAIM

- MISSION TARGET → intended future production scale
- SIGHT MANUFACTURING → electrolyser capability support

MECHANISM

- SIGHT PRODUCTION → hydrogen output support
- COMMISSIONED OUTPUT → later measured evidence

CONSEQUENCE / LIMIT

- CLIMATE OUTCOME → separate lifecycle result
- EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,...

MECHANISM / VERDICT — EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,...

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: State technology, unit,.

STAGE 11 ENERGY-TRANSITION ANSWER SPINE

ENERGY-TRANSITION ANSWER SPINE · CAPACITY GENERATION RELIABILITY SOURCE AND CARRIER · GRID INTEGRATION ELECTROLYSIS CERTIFICATION AND DELIVERY · DIRECT ELECTRIFICATION BEFORE SELECTIVE HYDROGEN USE · TARGET AWARD COMMISSION OUTPUT AND OUTCOME · DATED MNRE POWER-SYSTEM AND STANDARDS EVIDENCE

DEFINE → capacity generation reliability source and carrier → TRACE → grid integration electrolysis certification and delivery → SCREEN → direct electrification before selective hydrogen use → SEPARATE → target award commission output and outcome → VERIFY → dated MNRE power-system and standards evidence

CORE CLAIM

- DEFINE → capacity generation reliability source and carrier
- TRACE → grid integration electrolysis certification and delivery

MECHANISM

- SCREEN → direct electrification before selective hydrogen use
- SEPARATE → target award commission output and outcome

CONSEQUENCE / LIMIT

- VERIFY → dated MNRE power-system and standards evidence
- DEFINE → capacity generation reliability source and carrier

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated MNRE power-system and standards evidence.

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · LARGE-SCALE SOLAR AND WIND INFRASTRUCTURE REQUIRES SUBSTANTIAL LAND AREA · BALANCED FRAMING · THIS IS NOT A CASE WHERE EITHER RENEWABLE ENERGY · BATTERY-STORAGE COSTS HAVE BEEN FALLING BUT REMAIN A SUBSTANTIAL · ANALYTICAL POINT · INDIA'S RENEWABLE-CAPACITY-ADDITION TARGETS (500 GW NON-FOSSIL BY · GREEN-HYDROGEN PRODUCTION-COST FIGURES AND ELECTROLYSER-CAPACITY FIGURES CHANGE RAPIDLY

OPTIONAL SOURCE DEPTH

- CAUTION: Large-scale solar and wind infrastructure requires substantial land area and, for wind
- CAUTION: Balanced framing: this is not a case where either renewable energy or species
- CAUTION: Battery-storage costs have been falling but remain a substantial cost component of a
- CAUTION: Analytical point: India's renewable-capacity-addition targets (500 GW non-fossil by

ADVANCED DEBATE

- CAUTION: Green-hydrogen production-cost figures and electrolyser-capacity figures change rapidly
- CAUTION: Progress toward the National Green Hydrogen Mission's 5 MMT/year 2030 target and the
- CAUTION: Quantifying the precise trade-off between renewable-infrastructure expansion and
- Solar and wind generation are variable/non-dispatchable, requiring storage and grid

LEGACY / NUANCE

- Green hydrogen's cost-competitiveness depends on both falling electrolyser/renewable
- The Supreme Court has engaged with the tension between renewable-energy transmission
- The SIGHT Programme's dual incentive structure (electrolyser manufacturing plus
- NOT: India's renewable-energy challenge today is primarily about building enough generation

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels